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CSA0305

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – ANALYTICAL PROBLEM SOLVING

Course Code:	CSA0305	Course Title:	Data Structures
CO Assessed:	CO2 – Manipulation (BL3, BL4)	Assessment:	Assessment Tool 1 – Analytical Problem Solving
Weightage:	30% of CIA	Total Marks:	100
CO2 Statement:	Implement linear data structures such as stacks, queues, and linked lists, and analyze the efficiency of linear and binary search techniques.		
Student Name:	K. Gnaneswar Kumar	Reg. No.:	192572105
Section / Batch:	BE - CSE (AI)	Date:	20/07/2026

Code of Conduct

I K. Gnaneswar Kumar (192572105) (Name / Reg No)

certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: K. Gnaneswar Kumar

Instructions

- This test contains 5 Analytical problem-solving questions. Total: 100 Marks.
- When a student answer using an Analytical problem-solving, in evaluation the answer should be with following five requirements: Understanding of problem, select appropriate data structure, Design the solution, analyze, Clarity of representation.
- Each student should write 2 questions. No negative marking. Unanswered questions carry zero marks.
- No DTP printouts or electronic devices permitted. They should provide written materials.

Section / Topic	Focus	Q Nos	Marks
Linked Data Structure, Search Data Structure	List Operations, Binary Search	1	50
Stack Data Structure, Queue Data Structures	Stack Notations, Priority Queue	2	50
GRAND TOTAL:			100 Marks

Evaluate the following Postfix Expression using a stack
Expression: $2384+9-23*54*+9-$

Step	Symbol	Operation	Stack contents
1.	2	Push 2	2
2.	3	Push 3	2, 3
3.	*	Pop 3 and 2 $\rightarrow 2 \times 3 = 6$ $\rightarrow$ Push 6	6
4.	5	Push 5	6, 5
5.	4	Push 4	6, 5, 4
6.	*	Pop 4 and 5 $\rightarrow 5 \times 4 = 20$ $\rightarrow$ Push 20	6, 20
7.	+	Pop 20 and 6 $\rightarrow 6 + 20 = 26$ $\rightarrow$ Push 26	26
8.	9	Push 9	26, 9
9.	-	Pop 9 and 26 $\rightarrow 26 - 9 = 17$ $\rightarrow$ Push 17	17

* Result:-

Answer = 17 //

② Algorithm to Evaluate Any Postfix Expression using a stack.

*Algorithm:-

1. Start
2. Create an empty stack
3. Read the postfix expression from left to right.
4. If the symbol is an operand, push it onto the stack.
5. If the symbol is an operator:
 - * Pop the top two operands from the stack
 - * Perform the operation (Operand 1 operator Operand 2).
 - * Push the result back onto the stack.
6. Repeat steps 4 and 5 until the entire expression is scanned
7. The remaining element in the stack is the final answer.
8. Stop.

Pseudocode:

Algorithm EvaluatePostfix(Expression)

create an empty stack

For each symbol in Expression

 If symbol is an operand then

 Push(symbol)

 Else

 operand 2 = POP()

 operand 1 = POP()

 Result = operand 1 operator operand 2

 Push(Result)

END For

Answer = POP()

Print Answer

stop

* why are Stacks more suitable than Queues for Postfix Evaluation?

* Stack:

- Follows LIFO (Last In First Out) principle.
- The most recently inserted operands are immediately available for an operator.
- Supports Push and Pop operations efficiently.
- Makes Postfix Evaluation simple and efficient.

* Queue:

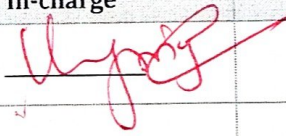
- Follows FIFO (First In First Out) principle.
- The first inserted operand is removed first, which is not suitable for postfix evaluation.
- Supports Enqueue and Dequeue operations, which do not match postfix evaluation requirements.
- Makes postfix evaluation difficult because operands are removed in the wrong order.

∴ Therefore, stacks are the most suitable data structure for Postfix Expression Evaluation.

38	192524251	VAISHNAV M	1. Two algorithms solve the same problem. Algorithm A uses recursion, while Algorithm B uses an explicit stack. Compare both approaches in terms of memory usage, execution flow, and risk of stack overflow. 2. A stack has a capacity of 5 elements. Explain what happens if the following operations are executed: Push(A), Push(B), Push(C), Push(D), Push(E), Push(F). Identify the error and suggest two possible solutions.
39	192525187	VEERA SANTHOSH A	1. Compare the performance of a linear queue and a circular queue when 100 enqueue and dequeue operations are performed alternately. 2. A queue implemented using an array report "Queue Full" even though some positions are empty. ➤ Analyze why this occurs. ➤ Explain how a circular queue solves this problem.
40	192511226	VINIL KUMAR S	1. Consider the singly linked list: 10 → 20 → 30 → 40 → 50. Delete the node containing 30. Explain how the links change after deletion. 2. Given two sorted linked lists: L1: 1 → 3 → 5 → 7 L2: 2 → 4 → 6 → 8

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding	20	Demonstrates complete and precise understanding; identifies all constraints and edge cases	Shows clear understanding with minor gaps	Partial understanding; misses some constraints	Misinterprets problem or lacks clarity
Data Structure Selection	20	Selects optimal data structure with strong justification and comparison	Selects appropriate structure with reasonable justification	Selects somewhat suitable structure with weak reasoning	Incorrect or no data structure selection
Algorithm Design	20	Designs efficient, logically sound, and well-structured algorithm	Algorithm is correct with minor inefficiencies	Algorithm is partially correct or lacks clarity	Algorithm is incorrect or missing
Accuracy of Solution	30	Error-free, well-structured, optimized code/solution	Minor errors but overall functional	Multiple errors; partially working	Non-functional or missing solution
Clarity	10	Exceptionally clear, well-organized, uses diagrams effectively	Clear and organized with minor issues	Somewhat organized but lacks clarity	Poorly organized and unclear

Faculty In-charge	Course Coordinator	Program Director
Signature: 	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____